

Bhanu Prasad Thota

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Professional Summary:

AI & Machine Learning Engineer with 6+ years of experience building, deploying, and optimizing AI/ML solutions across real estate, telecom, and research domains. Proven expertise in developing production-grade ML pipelines, LLM-powered agents, and multimodal AI applications using Python, TensorFlow, PyTorch, and cloud platforms like AWS SageMaker and Vertex AI. Adept at collaborating with cross-functional teams to deliver scalable solutions and drive successful adoption of AI technologies. Strong problem-solver with deep expertise in debugging, model optimization, and full-cycle ML system deployment.

Technical Skills:

- **Programming:** Python, SQL, Shell/Bash, C, IBM AS400
- **AI/ML Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LangChain, Transformers, MLflow, Airflow, FastAPI, Flask
- **Machine Learning & AI Expertise:** MLOps, Generative AI, LLMs (OpenAI, BERT-based, fine-tuning), Retrieval-Augmented Generation (RAG), Prompt Engineering, Computer Vision, Deep Learning, Model Deployment & Optimization, AI Troubleshooting & Debugging
- **Data Engineering & Storage:** PySpark, ETL Pipelines, Data Modeling, Data Governance, Snowflake, Pinecone, Vector Search, PostgreSQL, MySQL, NoSQL
- **Cloud Platforms:** AWS (SageMaker, EC2, S3, RDS, Lambda), GCP (Vertex AI), Azure (Data Explorer), Databricks
- **DevOps & Infrastructure:** Docker, Kubernetes, Kafka, GitHub Actions, CI/CD Pipelines
- **Collaboration & Agile Tools:** GitHub, Bitbucket, JIRA, Agile, SDLC, Customer-Facing Technical Support
- **Visualization & Reporting:** Matplotlib, Seaborn, Tableau, Excel
- **Operating Systems:** Linux, Windows

Professional Experience:

CentryStone

AI/ML Engineer

Norcross, GA

Feb 2025 – Present

- Designed and deployed scalable Spark-based preprocessing pipelines and FastAPI microservices to support real-time LLM chatbots and analytics workloads, processing thousands of concurrent customer requests with sub-second latency.
- Engineered LLM-powered conversational agents with long-term memory using Hugging Face and Pinecone vector DB, increasing response accuracy by 30% while improving customer engagement experience.
- Implemented and optimized RAG pipelines on cloud platforms (Vertex AI, SageMaker), reducing document retrieval latency by 40% through model quantization, caching strategies, and inference tuning.
- Partnered with cross-functional teams to troubleshoot deployment issues, debug model behaviors, and support clients during production rollout and onboarding phases.
- Automated customer support queries via voice-based AI bots, reducing frontline support workload by 70% and improving client satisfaction scores.
- Developed reusable Python packages and internal ML libraries with test-driven development practices, reducing post-deployment bugs by 40% and accelerating engineering velocity.

UMassD

Research & ML Engineer

Dartmouth, MA

Jan 2024 – Dec 2024

- Developed 3D human pose estimation pipelines using VIBE, YOLO, and SMPL-X models, enabling real-time mesh generation from monocular video streams.
- Built multimodal analytics pipelines combining CLIP vision-language embeddings and YOLO detections, improving action recognition accuracy by 25% on custom datasets.
- Optimized large-scale transformer models using selective neuron activation techniques, improving GPU memory utilization by 30% during training and inference.
- Engineered RNN-based sequence models for frame-wise video forecasting, boosting motion prediction accuracy by 40% in time-series video analysis.
- Designed an NLP-based SOP scoring engine using text embeddings, reducing manual admissions review effort by 50% while maintaining 85% prediction accuracy.
- Served as Teaching Assistant for Algorithms & Complexity courses, delivering lectures, leading recitation sessions, and guiding students through their academic projects.
- Collaborated with PhD researchers on cutting-edge AI projects, contributing to joint research efforts and expanding domain expertise through hands-on collaboration.

Netcracker Technology
Senior Software Engineer

Hyderabad, India
Apr 2022 – Dec 2022

- Built Spark-driven ETL pipelines and Python-SQL data validation frameworks for production analytics systems, reducing data cleansing and verification time by 60%.
- Partnered with solution architects to optimize Snowflake data models and reporting pipelines, improving BI dashboard load times by 35%.
- Developed internal CLI tools for ingestion validation and schema checks, decreasing manual post-deployment verification by 50%.
- Monitored Docker and Kubernetes clusters to proactively identify, troubleshoot, and resolve ETL memory issues, ensuring 99.9% production uptime.
- Contributed to technical design documentation, change request analysis, and functional specification reviews in collaboration with cross-functional business teams.

Software Engineer

Aug 2019 – Mar 2022

- Designed Flask-based REST APIs and Spark ETL pipelines for telecom data aggregation systems, improving response times by 15% and enhancing backend data integration.
- Led refactoring efforts to migrate legacy monolithic applications into Dockerized microservices, improving deployment flexibility, scalability, and fault isolation.
- Re-engineered CI/CD pipelines using GitHub Actions and Jenkins, streamlining build, test, and deployment processes and saving over 350+ hours annually in release cycles.
- Integrated Random Forest models into customer churn prediction pipelines, helping business units improve customer retention efforts by 12%.
- Acted as post-production support engineer, collaborating directly with business teams, stakeholders, and L2 support teams to triage, analyze, and resolve production incidents and system bugs while consistently meeting SLA timelines.
- Debugged multi-tier environments (API, database, ETL layers) to identify root causes of production outages and support rapid resolution.
- Participated in client review meetings to explain technical designs, clarify feature implementations, and address operational queries.

Associate Software Engineer

Jul 2018 – Jul 2019

- Crafted modular Python-SQL backend components to enhance telecom billing systems, increasing code reusability and accelerating feature rollout timelines.
- Automated recurring reporting and reconciliation processes with Python scripts, eliminating repetitive manual tasks and saving 8+ hours weekly.
- Maintained critical production systems with 95%+ uptime, actively monitoring system health, logs, and resolving incidents within agreed SLA windows.

Education:

University of Massachusetts Dartmouth

Dartmouth, MA

Master of Science in Data Science; GPA: 3.97

2024

Vasavi College of Engineering (Osmania University)

Hyderabad, India

Bachelor of Engineering in Mechanical Engineering; GPA: 3.53

2018

Achievements:

- **2nd Prize - Corsaire Idea Challenge (Charlton College of Business, UMass Dartmouth):** Awarded for building Resumate, an AI-powered resume optimization platform.
- **Donald C. Howard Pillars of the Community Leadership Award (2024):** Recognized for outstanding leadership and graduate community impact at UMass Dartmouth.
- **Star Performer - Netcracker Technologies:** Honored for high-impact technical contributions and delivery excellence.
- **ASME Human Powered Vehicle Challenge (2017):** Led national student team to Top 3 finish in endurance racing competition organized by the American Society of Mechanical Engineers (India Section).

Leadership & Activities:

- **Graduate Student Senate Senator (UMass Dartmouth, 2023 & 2024):** Twice-elected representative for the College of Engineering; led student advocacy, professional development programs, and graduate community engagement.
- **Student Chair – ASME Student Chapter:** Organized workshops, technical fests, and events for 900+ students, fostering innovation and academic collaboration.

- **Choreographer & Lead Dancer – College Dance Club:** Led and choreographed 15+ performances at university cultural events, contributing to campus diversity and community spirit.

Publications & Research:

- **Bhanu Prasad Thota, et al. *Multi-View Fusion for 3D Human Mesh Estimation Using Synthetic Views*** — Ongoing research focused on improving 3D human mesh estimation through synthetic view generation and fusion techniques.
- **Bhanu Prasad Thota, et al. *Accurating Human Video LLM by Eliminating Background Noise and Leveraging 3D Human Mesh*** — Ongoing research focused on enhancing video understanding LLMs using clean 3D mesh data to minimize visual noise and improve temporal reasoning.
- **Bhanu Prasad Thota, et al. *Conversion of Four Stroke IC Engine to Compressed Air Engine*** — Published and presented at the 6th National Conference on Advances in Mechanical Engineering (AIM-2016), contributing to sustainable mechanical engineering solutions. ISBN: 978-93-82570-82-0.